

MTH301 - Endsem

1. Define the Euclidean Inner product on $\mathbb{R}^n$. Prove the Cauchy-Schwarz Inequality. Also show that $\mathbb{R}^n$ is complete. (1 + 2 + 2 = 5)
2. Define a connected subset and a path connected subset of (X, d) . Show that a path connected subset is connected. Deduce that $\mathbb{R}^2$ is connected. (1 + 1 + 2 + 1 = 5)

(OR)

Consider a metric space (X, d) and fix $a \in X$. For each point $p \in X$, define $f_p : X \rightarrow \mathbb{R}$ by $f_p(x) = d(p, x) - d(a, x)$ for each $x \in X$. Show that f_p is uniformly continuous on X . Further, show that $\|f_p - f_q\|_\infty = d(p, q)$. (3 + 2 = 5)

3. Let $-\infty \leq a < b \leq \infty$, and $f : (a, b) \rightarrow \mathbb{R}$. Define the oscillation $\omega_f(x)$ of f at a point $x \in (a, b)$. Show that f is continuous at x if and only if $\omega_f(x) = 0$. Let g be the Thomae function on $\mathbb{R}$. Determine $\omega_g(x)$ for $x \in \mathbb{R}$. (1 + 2 + 2 = 5)
4. Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Show that $f \in \mathcal{R}[a, b]$. State the Riemann-Lebesgue Theorem. Deduce that the Thomae function g on $[0, 1]$ is Riemann Integrable on $[0, 1]$, i.e. $g \in \mathcal{R}[0, 1]$. (3 + 1 + 1 = 5)

(OR)

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function with continuous partial derivatives $\frac{\partial f}{\partial x}(x, y)$ and $\frac{\partial f}{\partial y}(x, y)$. Show that f is continuously differentiable. Suppose f has a local maximum at a point $(a, b) \in \mathbb{R}^2$. Then, show that $f'(a, b) = 0$. (3 + 2 = 5)

5. Let (X, d) be a compact metric space and $C(X)$ be the metric space with the sup-metric. Let $\mathcal{E}$ be a compact subset of $C(X)$. Show that $\mathcal{E}$ is equicontinuous. Is the subset $\mathcal{A} = \{f \in C[0, 1] : \|f\|_\infty \leq 1\}$ of $C[0, 1]$ compact? Explain. (3 + 2 = 5)
6. Say $(f_n)_{n \in \mathbb{N}}$ is a sequence of functions over $\mathcal{R}[a, b]$. Suppose (f_n) converges to f uniformly on $[a, b]$. Show that $f \in \mathcal{R}[a, b]$ and that $\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n$. (5)
7. State Weierstrass Approximation Theorem. Describe the Bernstein Polynomial of degree $n \geq 1$ associated to $f \in C[0, 1]$. Let $\phi \in C[-1, 1]$ with $\phi(0) = 0$ and $\int_{-1}^1 \phi(x) x^n dx = 0$ for all $n \in \mathbb{N}$. Show that $\phi(x) = 0$ for all $x \in [-1, 1]$. (1 + 1 + 3 = 5)
8. Define a contraction $\varphi : X \rightarrow X$. State Banach Contraction Principle. Let $X = \left[\sqrt{\frac{3}{2}}, \infty \right)$ be the closed interval of the real line and $f : X \rightarrow \mathbb{R}$ be given by

$$f(t) = \frac{1}{2} \left(t + \frac{3}{t} \right) \text{ for } t \in X$$

Show that the function f induces a contraction on X . Also, determine the unique fixed point of f . ($1 + 1 + 2 + 1 = 5$)

9. Let $E \subseteq \mathbb{R}^n$ be an open set and $f : E \rightarrow \mathbb{R}^m$ be differentiable at $a \in E$. Show that the total derivative $f'(a)$ is unique. Let $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ be the norm function $\psi(x) = \|x\|$ on $\mathbb{R}^n$. Let G be the set of all points $x \in \mathbb{R}^n$ such that $\psi'(x)$ exists. Determine G and, for each $x \in G$, determine the linear transformation $\psi'(x)$ giving all details. ($2 + 1 + 2 = 5$)
10. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function and let $v = (v_1, v_2) \in \mathbb{R}^2$ be a unit vector. Define the directional derivative $(D_v f)(x, y)$ of f at (x, y) in the direction of v . If f is differentiable at $(x, y) \in \mathbb{R}^2$, then, show that

$$(D_v f)(x, y) = v_1 \frac{\partial f}{\partial x}(x, y) + v_2 \frac{\partial f}{\partial y}(x, y)$$

Let $\phi(x, y) = \begin{cases} \frac{xy^2}{x^2+y^4} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$. Determine the directional derivative $(D_v \phi)(0, 0)$. ($1 + 2 + 2 = 5$)

(OR)

State the implicit function theorem. Let $f : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the mapping

$$f(x, y) = (x_1 x_2 - x_2^2 + y_1^2 - y_2^2, x_1 x_2 + x_1^2 + y_1^2 + y_2^2 - 4)$$

for $x = (x_1, x_2)$ and $y = (y_1, y_2) \in \mathbb{R}^2$. Determine the Jacobian matrix $J_f(x, y)$ of f at $(x, y) \in \mathbb{R}^2 \times \mathbb{R}^2$. Show that there is an open set $W \subseteq \mathbb{R}^2$ with $(1, 1) \in W$ and a continuously differentiable mapping $g : W \rightarrow \mathbb{R}^2$ such that $g((1, 1)) = (1, 1)$ and $f(g((y_1, y_2)), (y_1, y_2)) = 0$ for all $(y_1, y_2) \in W$. Determine the partial derivatives $\frac{\partial g}{\partial x}(1, 1)$ and $\frac{\partial g}{\partial y}(1, 1)$. ($1 + 1 + 2 + 1 = 5$)